



ZIRCON



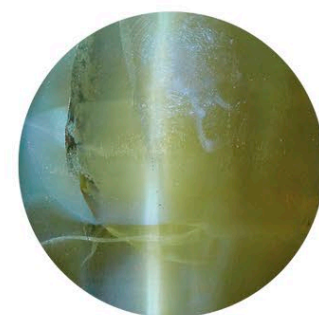
SPINEL



BARYTE



TURQUOISE



CAT'S EYE



TANZANITE



QUARTZ



AMETHYST



CHRYSOBERYL



APATITE



CALCITE

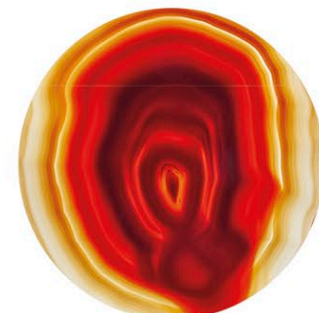
Optical variations Each gem has a wide-ranging set of optical properties, both in terms of colour and of clarity. During identification the gemologist has to take these into account, as well as the gem's refractive index (RI), its specific gravity (SG), hardness, lustre, and dichroism.



RUBY



CHALCEDONY



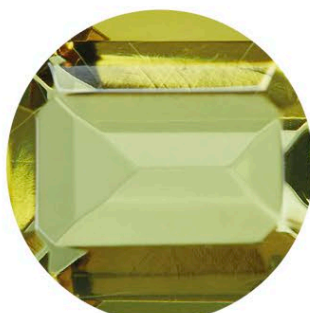
AGATE



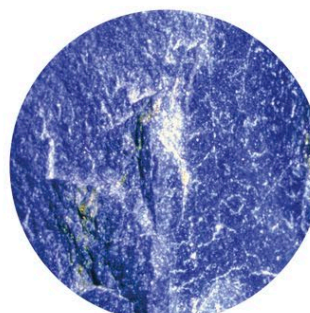
AMMONITE



JADEITE



AMBER



LAZURITE



COPAL

Visual properties

How a gem interacts with light is the very essence of its nature. Light is the source of a gemstone's beauty, colour, and sparkle; it is also a useful tool for the identification of gems, as each stone has its own particular set of optical properties. For example, there are a dozen or more red gemstones, and many of the red hues within each type of stone will have many different shades. All these properties are a way of identifying gemstones, although no single one is diagnostic in itself. Some

categories such as lustre are subjective observations; others, such as a mineral's refractive index, are objective. A gemologist identifying a stone will use a number of different methods and instruments to narrow the possibilities. Examination of one or all of the optical properties of the stone will show how it transmits, bends, and reflects light – just one of these may suffice for identification; in other instances, a complex combination of physical and optical properties may be needed.